

# A Lorentz-invariant background fixes the state-route constant to the radiation unit

**Result, 2026-09-27.** STATE item 2 asked for a physical reason why the constant  $\zeta$  of the state route (GPT-6 Astra's routes note, floor with  $h_* = 2\zeta$ ; the fifth-postulate note, Theorem B') should equal a fixed multiple of the radiation unit  $h_{\text{rad}}$  of Theorem U (unit-and-indeterminacy note). Within stochastic electrodynamics it does:

$$2\zeta = \frac{h_{\text{rad}}}{2\pi}, \quad \text{so} \quad h_* = \frac{h_{\text{rad}}}{2\pi} = \hbar.$$

Three steps. (1) A stationary random electromagnetic background that is Lorentz invariant has energy  $\kappa\omega$  per normal mode for one constant  $\kappa$ . (2) Every weakly damped linear oscillator in equilibrium with it has a Gaussian stationary state with  $\sqrt{\det \Sigma} = \kappa$ , whatever its mass and frequency (Theorem 1): the background realizes the covariant, noise-closed Gaussian restriction of Theorem B' with  $\zeta = \kappa$ , and it does so universally. (3) With the same background, Boyer derives the Planck spectrum from classical electrodynamics and classical thermodynamics, with zero-point energy  $\frac{1}{2}\hbar\omega$  per mode; so  $\kappa = \hbar/2$  and  $h_{\text{rad}} = 2\pi\hbar = 4\pi\kappa$ . The pure number of Astra's matching condition  $2\zeta = \gamma h_{\text{rad}}$  is therefore  $\gamma = 1/(2\pi)$  in this theory.

What stays open is Astra's closure premise: that no record, however it couples to the background, can infer the background's value and so sharpen the body's state below  $\zeta$ . Step (2) fixes  $\zeta$  for states in equilibrium with the background; it does not show that records cannot leave that set. The ingredients are established (Lorentz-invariant random radiation, Planck's resonator relation, linear response); the assembly is this note's, with no novelty claimed for the parts.

## 1. The background and the oscillator

**(1) Lorentz invariance fixes the spectrum up to one constant.** A homogeneous, isotropic, stationary random electromagnetic field whose spectrum is Lorentz invariant has energy density proportional to  $\omega^3$ , that is, energy per normal mode linear in frequency,  $\kappa\omega$  (Marshall 1963, metadata; Boyer 1969, abstract: "linear in frequency",  $\frac{1}{2}\hbar\omega$  per normal mode). The constant  $\kappa$  has the units of action, and Lorentz invariance leaves it free.

**Theorem 1 (universal Gaussian area).** Let a particle of mass  $M$  and charge  $e$  be bound harmonically at frequency  $\omega_0$ , with radiation damping small compared with  $\omega_0$ , and driven by the background of (1), taken Gaussian. Its stationary phase-space state is Gaussian with

$$\langle q^2 \rangle = \frac{\kappa}{M\omega_0}, \quad \langle p^2 \rangle = \kappa M\omega_0, \quad \langle qp + pq \rangle = 0, \quad \sqrt{\det \Sigma} = \kappa,$$

independent of  $M$ ,  $e$  and  $\omega_0$ .

*Proof.* The equation of motion is linear with Gaussian forcing, so the stationary state is Gaussian. A weakly damped oscillator in equilibrium with a radiation field acquires mean energy equal to the field's energy per normal mode at its frequency (Planck's resonator relation, derived in classical electrodynamics; Planck 1900, metadata), here  $\kappa\omega_0$ . For a harmonic oscillator the mean kinetic and potential energies are equal, so  $\langle p^2 \rangle / (2M) = \frac{1}{2}M\omega_0^2 \langle q^2 \rangle = \frac{1}{2}\kappa\omega_0$ , and stationarity gives  $\langle qp + pq \rangle = 0$ . Then  $\det \Sigma = \langle q^2 \rangle \langle p^2 \rangle = \kappa^2$ .  $\square$

The mass and the frequency drop out because the energy per mode is proportional to  $\omega$ : the area  $\langle E \rangle / \omega_0$  is the same for every oscillator. That is the universality the dimensional note requires of a floor (its Theorem A), supplied here by Lorentz invariance of the background rather than assumed.

**(2) The link to Theorem B'.** The set of Gaussian states with  $\sqrt{\det \Sigma} \geq \kappa$  is invariant under the affine symplectic group and closed under adding noise, and Theorem 1 says the background puts every harmonically bound body on its lower boundary. So the background

realizes the restriction of Theorem B' with  $\zeta = \kappa$ , and Astra's conditional floor applies with  $h_* = 2\kappa$  to comparisons whose records are confined to this set.

**(3) The link to the radiation unit.** Boyer derives the Planck law “without the formalism of quantum theory” from this background, classical electrodynamics of dipole oscillators, and classical equipartition of the particles' kinetic energy (Boyer 1969, abstract). The zero-point part is  $\frac{1}{2}\hbar\omega$  per mode, so  $\kappa = \hbar/2$ ; the thermal part is Planck's, whose constant is the  $h_{\text{rad}} = 2\pi\hbar$  of Theorem U. Hence

$$2\zeta = 2\kappa = \hbar = \frac{h_{\text{rad}}}{2\pi}.$$

This rests on Boyer's derivation at abstract level; the thermodynamic steps of that derivation are not re-derived here.

## 2. Consequence for STATE

STATE item 2 asked for two things: a physical justification of closure under recording, and the identification  $2\zeta = \gamma h_{\text{rad}}$ . The second is supplied within stochastic electrodynamics, with  $\gamma = 1/(2\pi)$ : one Lorentz-invariant constant sets both the equilibrium spectrum of radiation and the phase-space area of every bound body. The first remains open. It is now the single premise between a classical, pre-quantum account (Maxwell, Lorentz invariance, thermodynamics, a random background) and the Planck paper's floor on recording the inertial-versus-parabola comparison: that records cannot read the background and so leave its equilibrium set.